

Maximum Likelihood Estimation (MLE)

Math 742

2026-01-31

Introduction

The first MLE lecture focuses on what MLEs are and why they make sense. Large-sample theory appears later. Some detailed reading on large sample theory is in the appendix.

Another way to estimate a population parameter is **Maximum Likelihood Estimation (MLE)**. Remember that **Method of Moments (MoM)** compresses the data into a few summaries (e.g., mean, variance, skewness).

MLE, in contrast, is more like a CT scanner. Instead of looking at a single shadow and trying to reconstruct the object, the CT scanner captures **millions of tiny beams** passing through the object from all directions. Each beam provides a small piece of information about how the object blocks or bends light.

- One beam might indicate a bump.
- Another beam might show a hollow spot.
- Another beam reveals a thin edge.
- Put together, these beams allow the scanner to reconstruct the entire 3D object.

In essence, the scanner searches for the object that makes the observed pattern of beams **most consistent**. That is, the object for which the observed data (all the beams) are most likely.

Bottom line:

- MoM matches a few coarse features of the distribution.
- MLE tries to match the entire shape of the distribution using all the information in the sample.

Basic Idea

Suppose we observe an iid sample $X_1, X_2, \dots, X_n$ from a population with PDF or PMF $f(x|\theta_1, \theta_2, \dots, \theta_k)$, the likelihood function is defined by,

$$L(\theta_1, \theta_2, \dots, \theta_k) | \mathbf{x} = \prod_{i=1}^n f(x_i | \theta_1, \theta_2, \dots, \theta_k)$$

The idea of Maximum Likelihood Estimation is to choose the parameter values $(\theta_1, \theta_2, \dots, \theta_k)$ that make the observed data $\mathbf{x}$ **most plausible** under the model. That is, the values that maximize this joint density $L(\theta_1, \theta_2, \dots, \theta_k) | x$.

Definition: For each sample point $\mathbf{x}$, let $\hat{\theta}(\mathbf{x})$ be a parameter value at which $L(\theta | \mathbf{x})$ obtains a maximum as a function of θ , when $\mathbf{x}$ is held fixed. A *maximum likelihood estimator* (MLE) of the parameter θ based on a sample $\mathbf{X}$ is $\hat{\theta}(\mathbf{X})$.

Often, we take the log of the likelihood function. And, because the Log function is strictly monotone on its domain $(0, \infty)$, maximizing the log likelihood function is equivalent to maximizing the likelihood function. Why would you do this? Answer: Logs turn products into sums and algebraically this is usually simpler.

Procedure for determining MLE estimators

1. Write down the likelihood function ($L(\theta|x)$)
2. Take the log of the likelihood function
3. Differentiate and set equal to zero and solve
4. Take second derivative to check for concavity (i.e. that it's actually a maximum)

Normal, $\sigma^2 = 1$

Let $X_1, X_2, \dots, X_n$ be iid $N(\mu, 1)$. Then,

$$L(\mu, 1)|x = \prod_{i=1}^n \frac{1}{(2\pi)^{1/2}} e^{-(x_i - \mu)^2/2} = (2\pi)^{-n/2} e^{-(1/2) \sum_{i=1}^n (x_i - \mu)^2}$$

In order to find the value of μ that maximizes the likelihood I will maximize the log likelihood.

$$\ell(\mu, 1)|x = \log \left[(2\pi)^{-n/2} e^{-(1/2) \sum_{i=1}^n (x_i - \mu)^2} \right]$$

$$\ell(\mu, 1)|x = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2$$

Next, we differentiate with respect to μ and set equal to zero.

$$\frac{d(\ell(\mu, 1)|x)}{d\mu} = \frac{d(-\frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2)}{d\mu} = \sum_{i=1}^n (x_i - \mu) = 0$$

$$\sum_{i=1}^n x_i - n\mu = 0$$

$$\sum_{i=1}^n x_i = n\mu$$

$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

The last step is to take the second derivative and make sure that it is concave down.

$$\frac{d(\ell(\mu, 1)|x)}{d\mu} = \frac{d}{d\mu} \left(\sum_{i=1}^n (x_i - \mu) \right) = \frac{d}{d\mu} \left(\sum_{i=1}^n x_i - n\mu \right) = -n < 0$$

This means the function is concave down at the critical point so it is in fact a maximum.

Bernoulli

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$, $X_i \in \{0, 1\}$. The PMF for this distribution is,

$$f(x_1, x_2, \dots, x_n | p) = p^{x_i} (1 - p)^{1-x_i}$$

The likelihood function is:

$$L(p|x_i) = f(x_1, \dots, x_n | p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}, \quad x_i \in \{0, 1\}.$$

So, the log likelihood function is:

$$\ell(p|x_1, x_2, \dots, x_n) = \sum_{i=1}^n [x_i \log p + (1 - x_i) \log(1 - p)]$$

Let $S = \sum x_i$ then,

$$\ell(p|x_i) = S \log p + n \log(1 - p) - S \log(1 - p) = S \log p + (n - S) \log(1 - p)$$

Differentiate with respect to p and set equal to zero,

$$\frac{d(\ell(p|x_i))}{dp} = \frac{S}{p} - \frac{n - S}{1 - p} = 0$$

$$S(1 - p) = (n - S)p$$

$$S - Sp = np - Sp \implies S = np$$

$$\hat{p}_{MLE} = \frac{S}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Take the second derivative to verify that we have a maximum.

$$\begin{aligned} & \frac{d}{dp} \left(\frac{S}{p} - \frac{n - S}{1 - p} \right) \\ &= \frac{-S}{p^2} - \frac{n - S}{(1 - p)^2} \\ &= \frac{-S(1 - p)^2 - p^2(n - S)}{p^2(1 - p)^2} \end{aligned}$$

Note that for Bernoulli random variables, $S < n$, so the numerator is negative and the denominator is positive so this is concave down and hence a maximum.

Technically, $0 \leq S \leq n$, so what about the endpoints of the interval (i.e. $S = 0$ or $S = n$)? Well, $S = 0$ when $p = 0$ and $S = n$ when $p = 1$, otherwise (i.e. $p \in (0, 1)$) $0 < S < n$. Bottom line, the log likelihood is concave down if $p \in (0, 1)$.

What if $p = 0$ or $p = 1$ (degenerate case)?

- $S = 0$ every trial is a failure, $p = 0$.
- $S = n$ every trial is a success, $p = 1$

- Otherwise, $0 < S < n$

In each of these cases, the MLE still exists, it just lies on the boundary.

If $0 < p < 1$, then typically $0 < S < n$, and only in the extreme cases $S = n$ does the MLE lie on the boundary.

In practice, when $S = 0$ or $S = n$, we usually report $\hat{p} = 0$ or $\hat{p} = 1$, or use a continuity correction or Bayesian prior to avoid boundary estimates.

So, it's a maximum.

Binomial

Let $S = \sum_{i=1}^n X_i$ and $\hat{p} = S/n$. For $0 < p < 1$,

$$\ell(p) = S \log p + (n - S) \log(1 - p).$$

Then

$$\ell(p) - \ell(\hat{p}) = S \log\left(\frac{p}{\hat{p}}\right) + (n - S) \log\left(\frac{1 - p}{1 - \hat{p}}\right) = n \left[\hat{p} \log\left(\frac{p}{\hat{p}}\right) + (1 - \hat{p}) \log\left(\frac{1 - p}{1 - \hat{p}}\right) \right].$$

Since log is concave and $\hat{p} + (1 - \hat{p}) = 1$, Jensen's inequality implies

$$\hat{p} \log\left(\frac{p}{\hat{p}}\right) + (1 - \hat{p}) \log\left(\frac{1 - p}{1 - \hat{p}}\right) \leq \log\left(\hat{p} \frac{p}{\hat{p}} + (1 - \hat{p}) \frac{1 - p}{1 - \hat{p}}\right) = \log(1) = 0.$$

Hence $\ell(p) - \ell(\hat{p}) \leq 0$ (i.e. $\ell(p) \leq \ell(\hat{p})$) for all $p \in (0, 1)$, so $\hat{p} = S/n$ is a global maximizer of $\ell(p)$ on $(0, 1)$.

If $0 < S < n$, equality holds only when $p = \hat{p}$, so the maximizer is unique.

Poisson

Let $X_1, X_2, \dots, X_n \sim \text{Poisson}(\lambda)$. The PMF for this distribution is,

$$f(x_i | \lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$\ell(\lambda | x_1, x_2, \dots, x_n) = \sum_{i=1}^n x_i \log(\lambda) - n\lambda - \sum_{i=1}^n \log(x_i!)$$

Differentiate with respect to λ , set equal to zero.

$$\frac{1}{\lambda} \sum_{i=1}^n x_i - n - 0 = 0$$

$$\frac{1}{\lambda} \sum_{i=1}^n x_i = n$$

$$\hat{\lambda}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

Take the second derivative to ensure this is a maximum.

$$\frac{d}{d\lambda} \left(\frac{1}{\lambda} \sum_{i=1}^n x_i - n \right) = \frac{-1}{\lambda^2} \sum_{i=1}^n x_i < 0$$

So it's a maximum.

Invariance Property of the MLE

Suppose $\hat{\theta}$ is the MLE of θ , and let $\phi = g(\theta)$ be any one-to-one function of the parameter. Then the MLE of ϕ is

$$\hat{\phi} = g(\hat{\theta}).$$

This is called the **invariance property** of maximum likelihood estimators.

Example (Poisson). If $X \sim \text{Poisson}(\lambda)$, then $\mathbb{P}(X = 0) = e^{-\lambda}$. Since $\hat{\lambda} = \bar{X}$ is the MLE of λ , the MLE of $\mathbb{P}(X = 0)$ is

$$\mathbb{P}(\widehat{X = 0}) = e^{-\hat{\lambda}}.$$

Why this works:

MLE chooses the parameter value that makes the observed data most likely.

Any transformation of the parameter that preserves ordering of likelihood values must therefore be maximized at the transformed MLE.

Normal(μ, σ^2)

$$L(\mu, \sigma^2) | \mathbf{x} = \prod_{i=1}^n \frac{1}{(2\pi\sigma^2)^{1/2}} e^{-(x_i - \mu)^2 / (2\sigma^2)} = (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right)$$

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Now, differentiate with respect to μ ,

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2} 2 \sum_{i=1}^n (x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0$$

$$\sum_{i=1}^n (x_i - \mu) = m\bar{x} - n\mu = 0$$

$$\hat{\mu} = \bar{x}$$

Now, differentiated with respect to σ^2 .

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2 = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2(\sigma^2)(\sigma^2)} = 0$$

Multiply through by $2(\sigma^2)^2$,

$$-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 = 0$$

$$\sigma^2 n = \sum_{i=1}^n (x_i - \mu)^2$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

Of note, the MLE estimator is biased. $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ is unbiased.

Now take the second derivative to see if it's a max.

$$\frac{\partial}{\partial \mu} \left(\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) \right) = -\frac{n}{\sigma^2} < 0$$

$$\frac{\partial}{\partial \sigma^2} \left(-\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2(\sigma^2)(\sigma^2)} \right) = \frac{n}{2(\sigma^2)^2} - \frac{\sum_{i=1}^n (x_i - \mu)^2}{(\sigma^2)^3}$$

Now plug in the MLE, $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

$$\begin{aligned} & \frac{n}{2(\sigma^2)^2} - \frac{\sum_{i=1}^n (x_i - \mu)^2}{(\sigma^2)^3} \\ &= \frac{n}{2(\hat{\sigma}^2)^2} - \frac{n\hat{\sigma}^2}{(\hat{\sigma}^2)^3} = \frac{n}{2(\hat{\sigma}^2)^2} - \frac{n}{(\hat{\sigma}^2)^2} = -\frac{n}{2(\hat{\sigma}^2)^2} < 0 \end{aligned}$$

We have one more “mixed” second derivative to take.

$$\frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu),$$

which equals 0 at $\hat{\mu} = \bar{x}$. Therefore the Hessian is negative definite at $(\hat{\mu}, \hat{\sigma}^2)$, so this critical point is a (local) maximum.

Summary Table

Distribution	MLE
Bernoulli(p)	$\hat{p} = \bar{X} = \frac{\sum X_i}{n}$
Binomial(m, p)	$\hat{p} = \frac{\sum X_i}{mn}$
Poisson(λ)	$\hat{\lambda} = \bar{X} = \frac{\sum X_i}{n}$
Normal($\mu, 1$)	$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$
Normal(μ, σ^2)	$\hat{\mu} = \bar{x}, \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

Large-Sample Theory for MLE (Broad Overview)

This is a short, non-technical version that will hopefully give you an intuitive, picture-first view of large sample behavior of Maximum Likelihood Estimators (MLEs).

Suppose we observe data $X_1, \dots, X_n$ and fit a model with parameter θ using MLE. The MLE is the parameter value that makes the observed data most plausible (maximizes the likelihood function).

Some basic ideas to get in your head:

- More data $\rightarrow$ the sampling distribution of the MLE and/or the likelihood function becomes more peaked because the variance is smaller. The likelihood tends to concentrate near the true parameter.
- More informative data $\rightarrow$ sharper peak. When the data contain more information about the parameter (for example, through larger sample size or stronger signal), the log-likelihood becomes more sharply curved near its maximum. Sharper peaks correspond to smaller uncertainty in the estimate.
- Likelihood measures relative plausibility, not probability. The likelihood is proportional to the probability of the observed data given a parameter value, but it is not a probability distribution over the parameter itself. The MLE is simply the parameter value that makes the observed data most plausible under the model.

The curves below are *stylized, concave log-likelihood-like functions* constructed to illustrate curvature near the maximum; they are not the exact likelihoods for these models.

```
library(ggplot2)
library(dplyr)

# -----
# Idealized concave log-likelihood-like curves, all peaking at theta = 3
# Purpose: compare curvature near the peak (illustrative, not exact likelihoods)
# -----

# Parameter grid (theta is a "mean-like" parameter on (0, 6])
theta_vals <- seq(0.01, 6, length.out = 400)

# Normal(mean=theta, known variance) "log-likelihood-like" curve
# Using a quadratic centered at 3 (proportional to log-lik for x=3, sigma^2=1, up to constants)
norm_ll <- -0.5 * (theta_vals - 3)^2

# Binomial "mean-like" parameterization: mean = theta with n fixed
# If Y ~ Binomial(n, p), then E[Y] = n p = theta ==> p = theta/n
# Choose n and k so the MLE for theta (i.e., k) is 3
n_binom <- 10
k_binom <- 3
binom_ll <- k_binom * log(theta_vals / n_binom) +
  (n_binom - k_binom) * log(1 - theta_vals / n_binom)

# Poisson log-likelihood (up to constants) with MLE at theta = sample mean = 3
pois_ll <- 3 * log(theta_vals) - theta_vals

# -----
# Combine + normalize (subtract max so curves overlay cleanly)
# -----
df <- bind_rows(
  data.frame(param = theta_vals, ll = norm_ll, dist = "Normal"),
  data.frame(param = theta_vals, ll = binom_ll, dist = "Binomial"),
  data.frame(param = theta_vals, ll = pois_ll, dist = "Poisson")
)
```

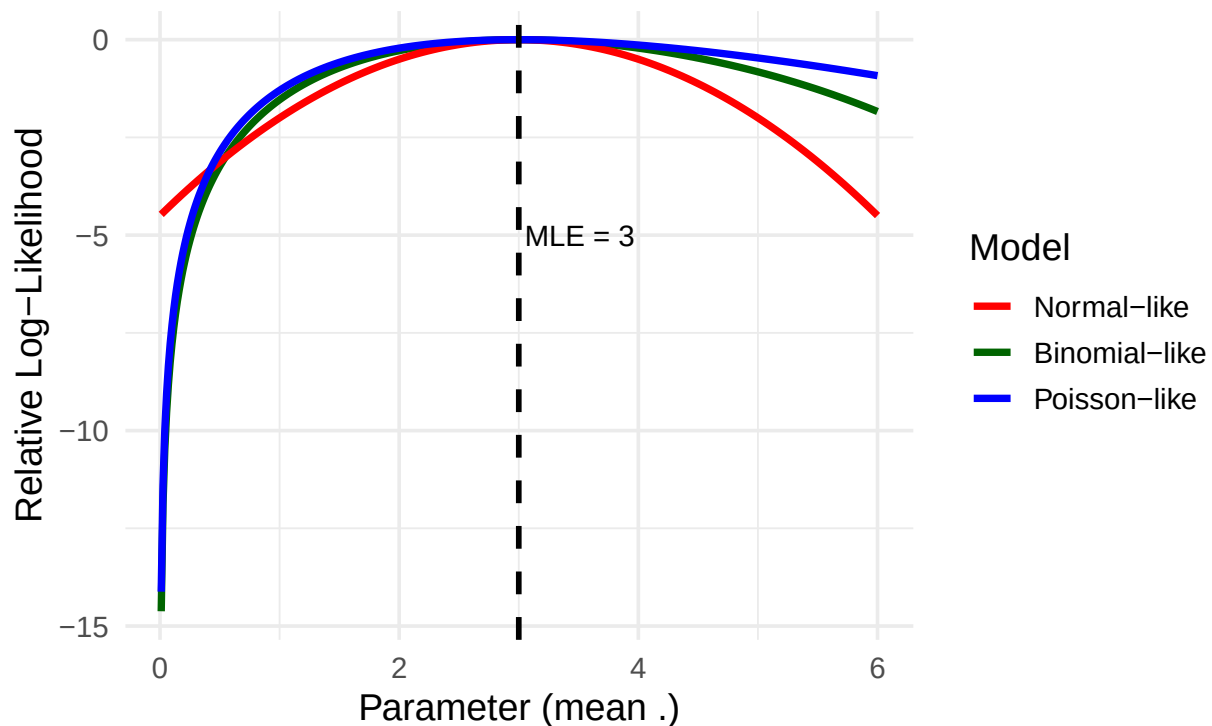
```

) %>%
  group_by(dist) %>%
  mutate(ll = ll - max(ll)) %>%
  ungroup()

# -----
# Plot
# -----
ggplot(df, aes(x = param, y = ll, color = dist)) +
  geom_line(linewidth = 1.3) +
  geom_vline(xintercept = 3, linetype = "dashed", linewidth = 1) +
  annotate("text", x = 3.05, y = -5, label = "MLE = 3", hjust = 0, size = 4) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Idealized Concave Log-Likelihood Shapes (Illustrative)\n(All Peak at Mean = 3)",
    x = "Parameter (mean \u03b8)",
    y = "Relative Log-Likelihood",
    color = "Model"
  ) +
  scale_color_manual(
    values = c(
      "Normal" = "red",
      "Binomial" = "darkgreen",
      "Poisson" = "blue"
    ),
    breaks = c("Normal", "Binomial", "Poisson"),
    labels = c("Normal-like", "Binomial-like", "Poisson-like")
  )

```


Idealized Concave Log-Likelihood Shapes (Illustrative) (All Peak at Mean = 3)



The figure shows the log-likelihood functions for three common distributions—Normal, Binomial, and Poisson—constructed so that all three peak at the same value ($\theta = 3$). Each curve rises to a maximum at its MLE and falls away on either side, illustrating a key principle of likelihood-based inference:

- The parameter value that makes the observed data most probable is the estimator.
- The steepness of each curve near its peak reflects the certainty of that estimate—sharper peaks correspond to higher information and lower standard error.

These shapes give a visual, intuitive sense for why the MLE is the maximizer of the log-likelihood and how likelihood curvature connects to uncertainty.

Consistency

Basic Idea: As we collect more data, the MLE tends to get closer and closer to the true parameter value (assuming the model is correctly specified and identifiable).

Consistency tells us the estimator is not hopeless, with enough data it finds the truth (in probability). It does not say how fast it converges or give a sampling distribution. $\hat{\theta} \xrightarrow{p} \theta_0$

```
set.seed(101)

library(ggplot2)
library(dplyr)

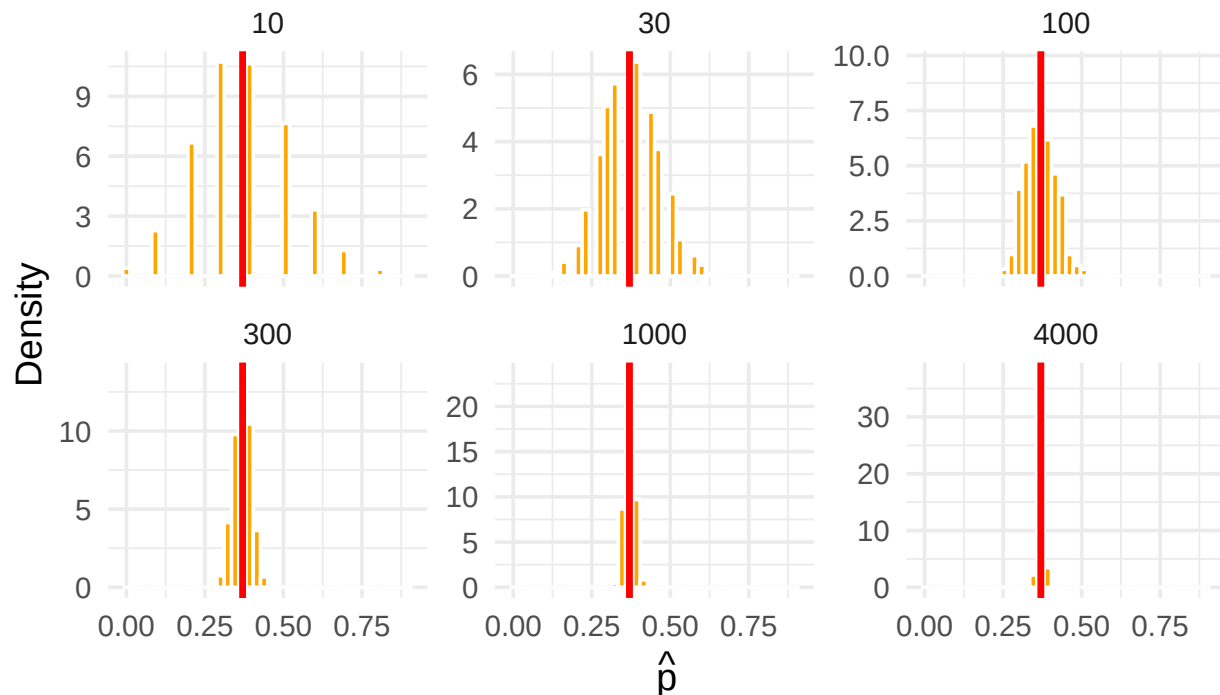
p_true <- 0.37
ns <- c(10, 30, 100, 300, 1000, 4000)
B <- 2000 # number of repeated experiments per n
```

```
df_cons <- lapply(ns, function(n) {
  phat <- replicate(B, mean(rbinom(n, 1, p_true)))
  data.frame(n = n, phat = phat)
}) |> bind_rows()

ggplot(df_cons, aes(x = phat)) +
  geom_histogram(aes(y = after_stat(density)),
    bins = 40,
    fill = "orange",
    color = "white") +
  geom_vline(xintercept = p_true, color = "red", linewidth = 1.2) +
  facet_wrap(~ n, scales = "free_y") +
  theme_minimal(base_size = 14) +
  labs(
    title = "Consistency of the MLE for Bernoulli(p)",
    subtitle = "As sample size increases, the sampling distribution of phat concentrates near the true p",
    x = expression(hat(p)),
    y = "Density"
  )
```

Consistency of the MLE for Bernoulli(p)

As sample size increases, the sampling distribution of phat concentrates near the true p



Asymptotic Normality

Basic Idea: For large sample sizes, the distribution of the MLE is approximately normal and centered at the true parameter, with spread shrinking at rate $1/\sqrt{n}$. Concretely, if $\hat{\theta}_n$ is the MLE then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}),$$

$I(\theta_0)$ is called the Fisher Information (covered later), where $I(\theta_0)$ is the Fisher information for a single observation (equivalently, the curvature of the log-likelihood at θ_0).

This explains why confidence intervals and hypothesis tests using normal approximations often work well for moderate sample sizes.

```
set.seed(1)
library(ggplot2)
library(patchwork)

p_true <- 0.25

# n = 50 -----
phat50 <- replicate(50000, mean(rbinom(50, 1, p_true)))

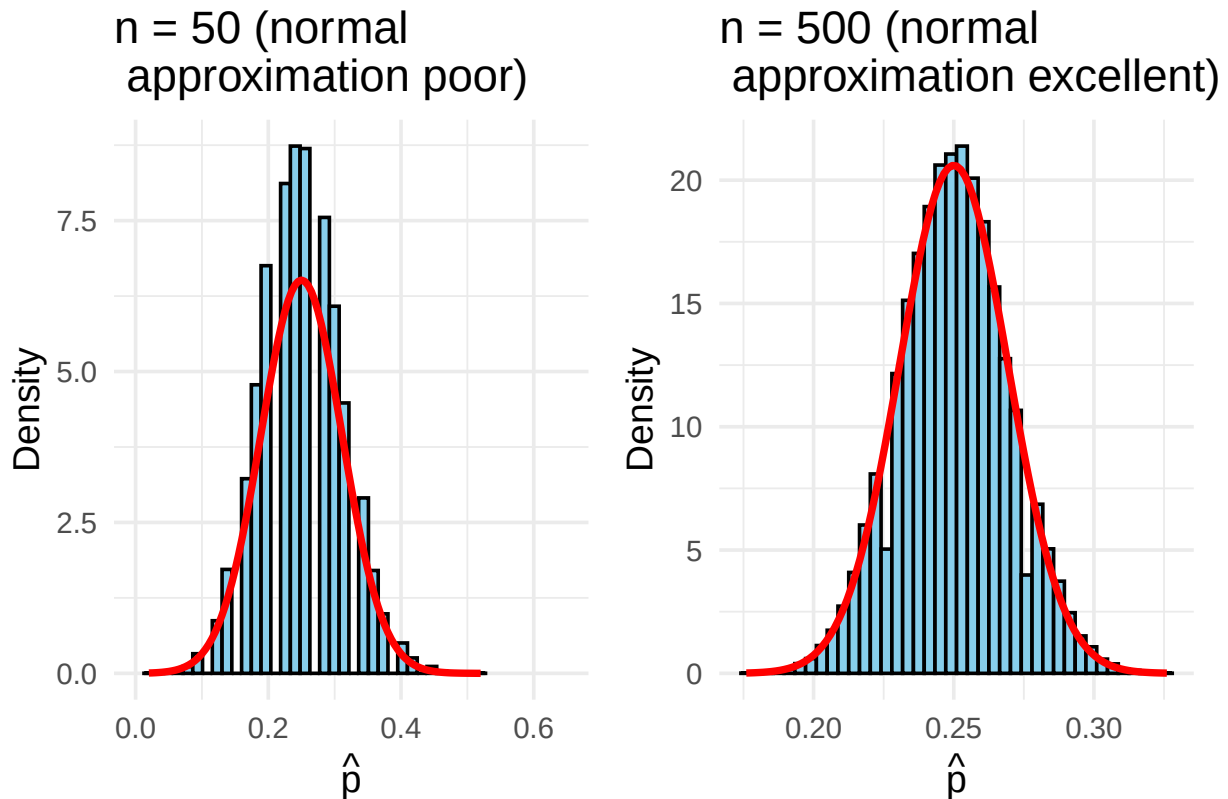
p50 <- ggplot(data.frame(phat = phat50), aes(x = phat)) +
  geom_histogram(aes(y = after_stat(density)), bins = 35, fill = "skyblue", color = "black") +
  stat_function(fun = dnorm,
               args = list(mean = p_true,
                           sd = sqrt(p_true*(1-p_true)/50)),
               color = "red", size = 1.3) +
  coord_cartesian(xlim = c(0, 0.65)) + # zoom in, cut off the empty right tail
  labs(title = "n = 50 (normal \n approximation poor)",
       x = expression(hat(p)), y = "Density") +
  theme_minimal(base_size = 14)

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

# n = 500 -----
phat500 <- replicate(50000, mean(rbinom(500, 1, p_true)))

p500 <- ggplot(data.frame(phat = phat500), aes(x = phat)) +
  geom_histogram(aes(y = after_stat(density)), bins = 40, fill = "skyblue", color = "black") +
  stat_function(fun = dnorm,
               args = list(mean = p_true,
                           sd = sqrt(p_true*(1-p_true)/500)),
               color = "red", size = 1.3) +
  labs(title = "n = 500 (normal \n approximation excellent)",
       x = expression(hat(p)), y = "Density") +
  theme_minimal(base_size = 14)

p50 | p500
```



For the Bernoulli proportion MLE $\hat{p} = \bar{X}$, asymptotic normality states that $\sqrt{n}(\hat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1 - p))$ as $n \rightarrow \infty$. When $p = 0.25$ and $n = 50$, the sampling distribution of $\hat{p}$ remains visibly right-skewed and bounded away from zero, so the normal approximation $\hat{p} \approx \mathcal{N}(p, p(1 - p)/n)$ is poor, the overlaid bell curve is too flat, too wide, and places impossible mass below 0. By $n = 500$, however, skewness has largely vanished, the distribution is nearly symmetric, and the normal curve with mean $p = 0.25$ and standard deviation $\sqrt{0.25 \times 0.75/n}$ matches the simulated histogram almost perfectly, illustrating that asymptotic normality is an excellent approximation once the sample size is sufficiently large, even for p away from 0.5.

Fisher Information

Basic Idea: Fisher information measures how much information the data carry about the parameter. It is connected to the curvature of the expected log-likelihood (averaged over repeated samples): a steep, sharply peaked expected log-likelihood $\implies$ high information.

If you have more information, MLE variance is smaller. In many models, the asymptotic variance of the MLE is approximately the inverse of the sample size times the Fisher information per observation.

In many models,

$$\mathbb{V}(\hat{\theta}) \approx \frac{1}{nI(\theta)}$$

- Bernoulli with p near 0.5 $\rightarrow$ more information per trial than p near 0.01 (because when p is extreme, outcomes are almost always the same and give little information about the exact value of p).
- For Normal with known variance, information about the mean is proportional to $1/\sigma^2$. Less noise $\rightarrow$ more information.

Standard Error $\approx$ Inverse Curvature at Peak

The curvature (second derivative) of the log-likelihood at its maximum measures how rapidly the likelihood falls off when we move away from the maximizer. A tight curvature means small uncertainty. Thus to get an approximate standard error we inspect the curvature.

- Picture flat vs sharp peaks and relate to wide vs narrow approximate sampling distributions.
- In the future, we'll compute curvature via derivatives (the Hessian) and obtain the commonly used inverse-Hessian standard errors. No calculus required now.

Appendix (Optional: Sketch for intuition (not examinable))

The more detailed information on Large-Sample Theory will be covered later

Large-Sample Theory for MLE (Detailed)

Let $X_1, \dots, X_n$ be iid observations from a density or pmf $f(x|\theta)$ indexed by a parameter $\theta \in \Theta \subseteq \mathbb{R}^k$. The likelihood and log-likelihood are

$$L(\theta_1, \theta_2, \dots, \theta_k | x) = \prod_{i=1}^n f(x_i | \theta_1, \theta_2, \dots, \theta_k)$$

$$\ell(\theta_1, \theta_2, \dots, \theta_k | x) = \sum_{i=1}^n \log f(x_i | \theta_1, \theta_2, \dots, \theta_k)$$

We treat the observed data as fixed and view the log-likelihood as a function of the parameter. The MLE is any maximizer (or sequence of maximizers) of $\ell(\theta)$:

Consistency (informal)

Statement (informal). Under suitable regularity conditions (identifiability, continuity of the log-likelihood, and that the population expected log-likelihood is uniquely maximized at the true parameter), any sequence of MLEs $\hat{\theta}_n$ converges in probability to the true parameter θ_0 . That is,

$$\hat{\theta}_n \xrightarrow{p} \theta_0.$$

Intuition / proof sketch. Define the average log-likelihood

$$\bar{\ell}_n(\theta) = n^{-1} \ell_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log f(X_i | \theta).$$

By the Law of Large Numbers, for each fixed θ ,

$$\bar{\ell}_n(\theta) \xrightarrow{p} \mathbb{E}_{\theta_0}[\log f(X | \theta)] =: M(\theta).$$

If $M(\theta)$ is uniquely maximized at θ_0 (identifiability condition) and the convergence is uniform in a neighborhood, then the maximizer of $\bar{\ell}_n(\theta)$ converges to the maximizer of $M(\theta)$, i.e. θ_0 . Thus the (sequence of) MLEs is consistent.

Practical note. Consistency does *not* give a convergence rate (but typically the MLE converges at rate $n^{-1/2}$ under regularity conditions — see next section).

Bottom Line: This is just the Law of Large Numbers + Continuous Mapping Theorem applied to the argmax.

Asymptotic normality (the classical result)

One-parameter version (scalar (θ)). Under standard regularity conditions, the MLE satisfies

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}),$$

where $I(\theta_0)$ is the Fisher information per observation (defined below). Equivalently,

$$\hat{\theta}_n \approx N(\theta_0, [nI(\theta_0)]^{-1}) \quad (\text{for large } n).$$

Multivariate version. For $\theta \in \mathbb{R}^k$,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N_k(0, I(\theta_0)^{-1})$$

with $I(\theta_0)$ the $k \times k$ Fisher information matrix.

Sketch of derivation (informal). Expand the score (first derivative of log-likelihood) around θ_0 :

$$0 = \dot{\ell}_n(\hat{\theta}_n) = \dot{\ell}_n(\theta_0) + \ddot{\ell}_n(\tilde{\theta}_n)(\hat{\theta}_n - \theta_0),$$

where $\tilde{\theta}_n$ lies between $\hat{\theta}_n$ and θ_0 . Rearranging,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = -\left[-\frac{1}{n}\ddot{\ell}_n(\tilde{\theta}_n)\right]^{-1} \frac{1}{\sqrt{n}}\dot{\ell}_n(\theta_0).$$

Under regularity, $n^{-1}\ddot{\ell}_n(\tilde{\theta}_n) \xrightarrow{P} \mathbb{E}[-\ddot{\ell}(X | \theta_0)] = I(\theta_0)$ and, by the Central Limit Theorem,

$$\frac{1}{\sqrt{n}}\dot{\ell}_n(\theta_0) \xrightarrow{d} N(0, I(\theta_0)).$$

Combining gives the stated asymptotic normality.

Fisher information

Definition (scalar). The Fisher information in a single observation about parameter θ is

$$I(\theta) = \mathbb{E} * \theta \left[\left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right)^2 \right] = -\mathbb{E} * \theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X | \theta) \right],$$

when the expectations exist and the regularity conditions hold that allow moving derivatives inside the expectation. For vector parameters, replace derivatives with gradients and the scalar with the information matrix

$$I(\theta) = \mathbb{E}[\dot{\ell}(X | \theta)\dot{\ell}(X | \theta)^\top] = -\mathbb{E}[\ddot{\ell}(X | \theta)].$$

Interpretation. Fisher information measures curvature of the expected log-likelihood (how rapidly the likelihood distinguishes nearby parameter values). Large information means the log-likelihood is sharply peaked and the MLE has small variance.

Examples (common models).

- Normal with known variance σ^2 : $X \sim N(\mu, \sigma^2)$, parameter μ .

$$I(\mu) = \frac{1}{\sigma^2}.$$

For n i.i.d. observations the information is n/σ^2 .

- Bernoulli(p): $I(p) = 1/[p(1-p)]$ (per observation). For n observations the information is $n/[p(1-p)]$.
- Poisson(λ): $I(\lambda) = 1/\lambda$ (per observation).

Approximating standard errors (observed information)

Given asymptotic normality, a natural approximate variance for $\hat{\theta}_n$ is

$$\widehat{\text{Var}}(\hat{\theta}_n) \approx [nI(\hat{\theta}_n)]^{-1}.$$

Two common approaches to estimate the information/variance:

1. **Expected information (plug-in).** Compute the Fisher information formula and plug in $\hat{\theta}_n$.
2. **Observed information (Hessian).** Use the observed negative second derivative of the log-likelihood at the MLE:

$$\mathcal{I}_{obs}(\hat{\theta}_n) = -\frac{1}{n}\ddot{\ell}_n(\hat{\theta}_n).$$

Then approximate

$$\widehat{\text{Var}}(\hat{\theta}_n) \approx [n, \mathcal{I}_{obs}(\hat{\theta}_n)]^{-1} = -[\ddot{\ell}_n(\hat{\theta}_n)]^{-1}.$$

Worked examples

Example A — Bernoulli

We observe $X_1, \dots, X_n \sim \text{Bernoulli}(p)$. The log-likelihood (dropping constants) is

$$\ell(p) = S \log p + (n - S) \log(1 - p), \quad S = \sum X_i.$$

The MLE is $\hat{p} = S/n$. The Fisher information (per observation) is $I(p) = 1/[p(1-p)]$, so the approximate variance is $[nI(p)]^{-1} = p(1-p)/n$.

When θ is scalar, the standard error is the square root of this variance estimate.

```
set.seed(999)
# simulate data
n <- 100
p_true <- 0.37
x <- rbinom(n, size=1, prob=p_true)
S <- sum(x)
phat <- S/n
# observed information (scalar) from second derivative of log-lik
# log-lik: S*log(p) + (n-S)*log(1-p)
# second derivative: -S/p^2 - (n-S)/(1-p)^2
obs_info <- -(-S/phat^2 - (n-S)/(1-phat)^2)/n # divide by n to get per-observation
se_plugin <- sqrt(phat*(1-phat)/n)
list(n=n, S=S, phat=phat, se_plugin=se_plugin, obs_info_per_obs=obs_info)
```

```
## $n
## [1] 100
##
## $S
## [1] 32
##
## $phat
## [1] 0.32
##
## $se_plugin
## [1] 0.04664762
##
## $obs_info_per_obs
## [1] 4.595588
```

Interpretation: 0.0466476 gives the usual standard error for a sample proportion. The observed information per observation should be close to the theoretical $1/[p(1-p)]$ when n is large.

Example B — Normal, unknown mean (known variance)

Assume $X_i \sim N(\mu, \sigma^2)$ with known $\sigma^2 = 1$. The log-likelihood (up to constants) is

$$\ell(\mu) = -\frac{1}{2} \sum (X_i - \mu)^2.$$

The MLE is $\hat{\mu} = \bar{X}$. The Fisher information (per observation) is $I(\mu) = 1/\sigma^2$; with $\sigma^2 = 1$ the information is 1 and $\text{Var}(\hat{\mu}) \approx 1/n$.

```
n <- 50
mu_true <- 1.2
sigma <- 1
x <- rnorm(n, mean=mu_true, sd=sigma)
# MLE for mu:
mu_hat <- mean(x)
# observed information: -1/n * second derivative; second derivative of log-lik is -n/sigma^2
obs_info_per_obs <- 1/sigma^2
se_hat <- sqrt(1/(n*obs_info_per_obs))
list(mu_hat=mu_hat, se_hat=se_hat, obs_info_per_obs=obs_info_per_obs)

## $mu_hat
## [1] 1.243281
##
## $se_hat
## [1] 0.1414214
##
## $obs_info_per_obs
## [1] 1

import numpy as np

# Parameters
np.random.seed(0) # optional, for reproducibility
n = 50
mu_true = 1.2
sigma = 1
```



```

# Simulate data
x = np.random.normal(mu_true, sigma, n)

# MLE for mu
mu_hat = np.mean(x)

# Observed information per observation
# log-lik:  $-(n/2) \cdot \log(2\pi \cdot \sigma^2) - \sum((x-\mu)^2)/(2\sigma^2)$ 
# second derivative w.r.t mu:  $-n/\sigma^2$ 
obs_info_per_obs = 1 / sigma**2

# Standard error of mu_hat
se_hat = np.sqrt(1 / (n * obs_info_per_obs))

# Results
print(f"mu_hat: {mu_hat:.4f}")

## mu_hat: 1.3406
print(f"se_hat: {se_hat:.4f}")

## se_hat: 0.1414
print(f"obs_info_per_obs: {obs_info_per_obs:.4f}")

## obs_info_per_obs: 1.0000

```

Small simulation to observe asymptotic normality

We simulate many MLEs for a Bernoulli and plot a histogram overlaid with the asymptotic normal approximation.

```

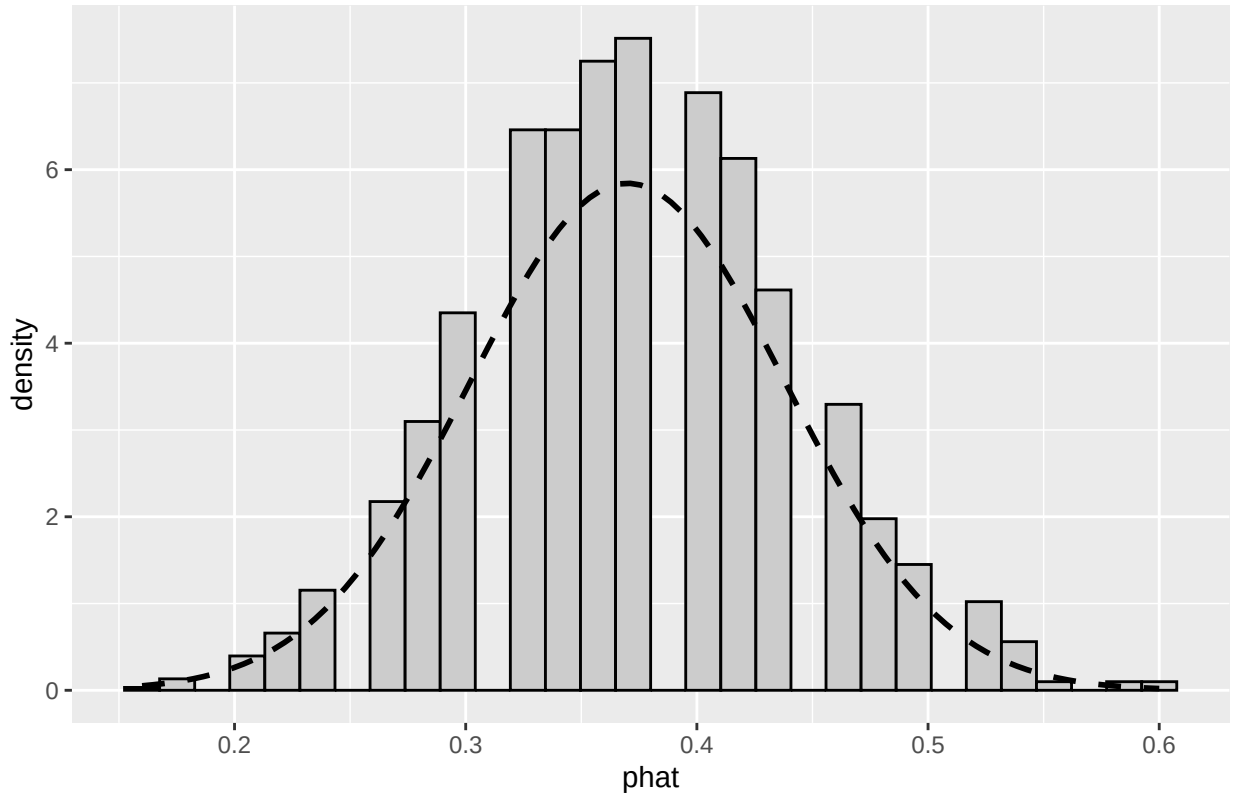
library(ggplot2)
set.seed(2)
n <- 50
p_true <- 0.37
B <- 2000
phats <- replicate(B, mean(rbinom(n,1,p_true)))
# approximate normal density
df <- data.frame(phat=phats)

p1 <- ggplot(df, aes(phat)) +
  geom_histogram(aes(y=..density..), bins=30, color='black', fill='grey80') +
  stat_function(fun = function(x) dnorm(x, mean=p_true, sd=sqrt(p_true*(1-p_true)/n)),
               size=1, linetype='dashed') +
  labs(title=sprintf('Distribution of MLE (n=%d, p=%.2f)', n, p_true), x='phat', y='density')

p1

```

Distribution of MLE (n=50, p=0.37)



Practical remarks and caveats

- **Regularity conditions matter.** The theorems require differentiability, identifiability, and interchange of expectation and differentiation. For irregular models (e.g. parameters on the boundary, non-identifiability, mixture models) the classical results may fail.

Rate of convergence. Under regularity, MLEs converge at the $(\hat{n}^{-1/2})$ rate. Some estimators (e.g. non-parametric or semi-parametric) may converge slower.

Observed vs expected information. Both provide asymptotically equivalent variance estimators; in finite samples they can differ. The observed information is convenient since it can be computed from the Hessian returned by numerical optimizers.

Delta method. To get standard errors for functions $g(\hat{\theta}_n)$, use the delta method: if $\sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow N(0, \Sigma)$, then

$$\sqrt{n}(g(\hat{\theta}_n) - g(\theta_0)) \rightarrow N(0, Dg(\theta_0)\Sigma Dg(\theta_0)^\top),$$

where Dg is the gradient/Jacobian of g at θ_0 .

Bottom line: For regular models and large n , the MLE is approximately normal with mean θ_0 and variance $1/(nI(\theta))$. This is why 95% CIs are routinely $\theta \pm 1.96$ SE.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
```

```

# -----
# Idealized concave log-likelihood-like curves, all peaking at theta = 3
# (Illustrative shapes, not exact likelihoods)
# -----

# Parameter grid
theta_vals = np.linspace(0.01, 6, 400)

# Normal(mean=theta, sigma^2=1) "log-likelihood-like" curve
# Quadratic centered at 3
norm_ll = -0.5 * (theta_vals - 3)**2

# Binomial "mean-like" parameterization
# Mean = n * p = theta ==> p = theta / n
n_binom = 10
k_binom = 3
binom_ll = (
    k_binom * np.log(theta_vals / n_binom)
    + (n_binom - k_binom) * np.log(1 - theta_vals / n_binom)
)

# Poisson log-likelihood (up to constants)
pois_ll = 3 * np.log(theta_vals) - theta_vals

# -----
# Normalize so curves overlay cleanly
# -----
norm_ll -= np.max(norm_ll)
binom_ll -= np.max(binom_ll)
pois_ll -= np.max(pois_ll)

# -----
# Plot
# -----
plt.figure(figsize=(9, 6))

plt.plot(theta_vals, norm_ll, color="red", linewidth=2, label="Normal-like")
plt.plot(theta_vals, binom_ll, color="darkgreen", linewidth=2, label="Binomial-like")
plt.plot(theta_vals, pois_ll, color="blue", linewidth=2, label="Poisson-like")

plt.axvline(3, color="black", linestyle="dashed", linewidth=1)
plt.text(3.05, -5, "MLE = 3", fontsize=11)

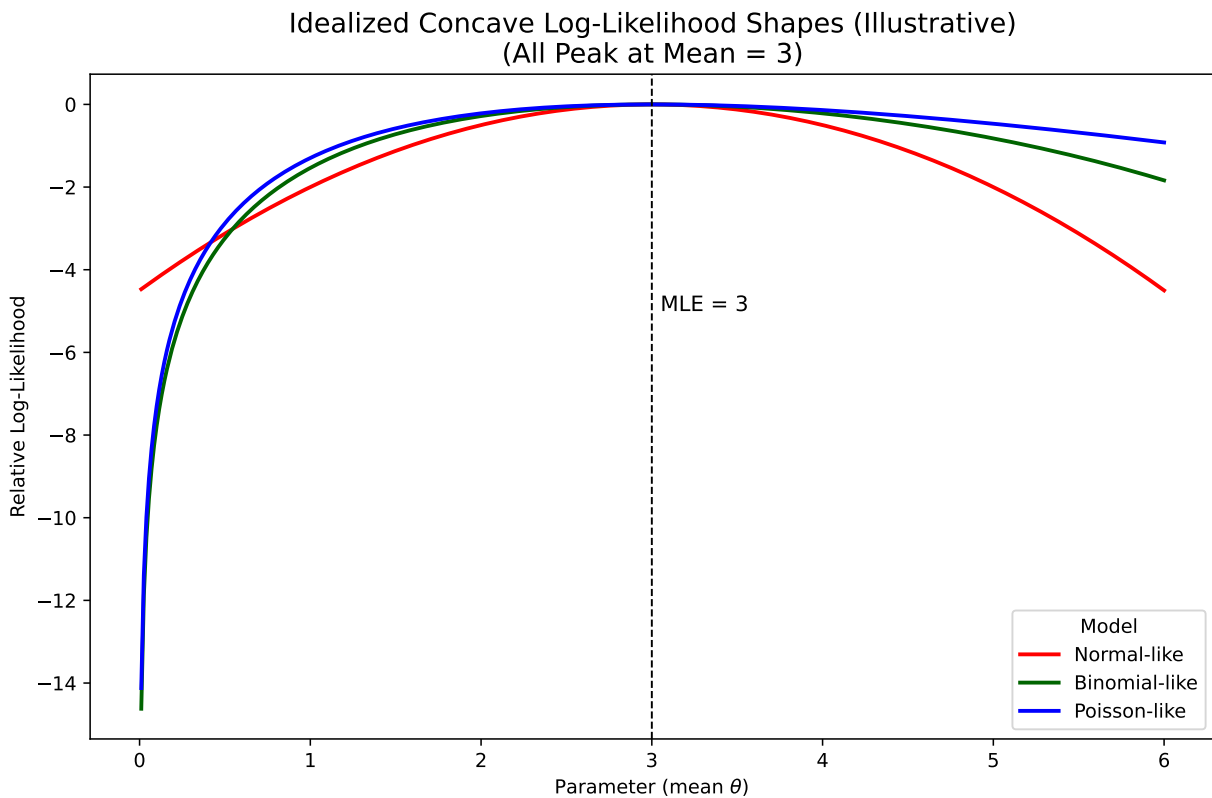
plt.title(
    "Idealized Concave Log-Likelihood Shapes (Illustrative)\n"
    "(All Peak at Mean = 3)",
    fontsize=14
)

plt.xlabel(r"Parameter (mean $\theta$)")
plt.ylabel("Relative Log-Likelihood")

plt.legend(title="Model")
plt.tight_layout()

```

```
plt.show()
```



```
import numpy as np
import matplotlib.pyplot as plt

# -----
# Simulation settings
# -----
np.random.seed(101)

p_true = 0.37
ns = [10, 30, 100, 300, 1000, 4000]
B = 2000 # repetitions per n

# -----
# Simulate MLEs
# -----
phats = {}
for n in ns:
    phats[n] = np.array([
        np.mean(np.random.binomial(1, p_true, size=n))
        for _ in range(B)
    ])

# -----
# Plot: faceted histograms
# -----
```

```

fig, axes = plt.subplots(2, 3, figsize=(14, 8), sharex=False, sharey=False)
axes = axes.flatten()

for ax, n in zip(axes, ns):
    data = phats[n]

    _ = ax.hist(
        data,
        bins=40,
        density=True,
        color="orange",
        edgecolor="white"
    )

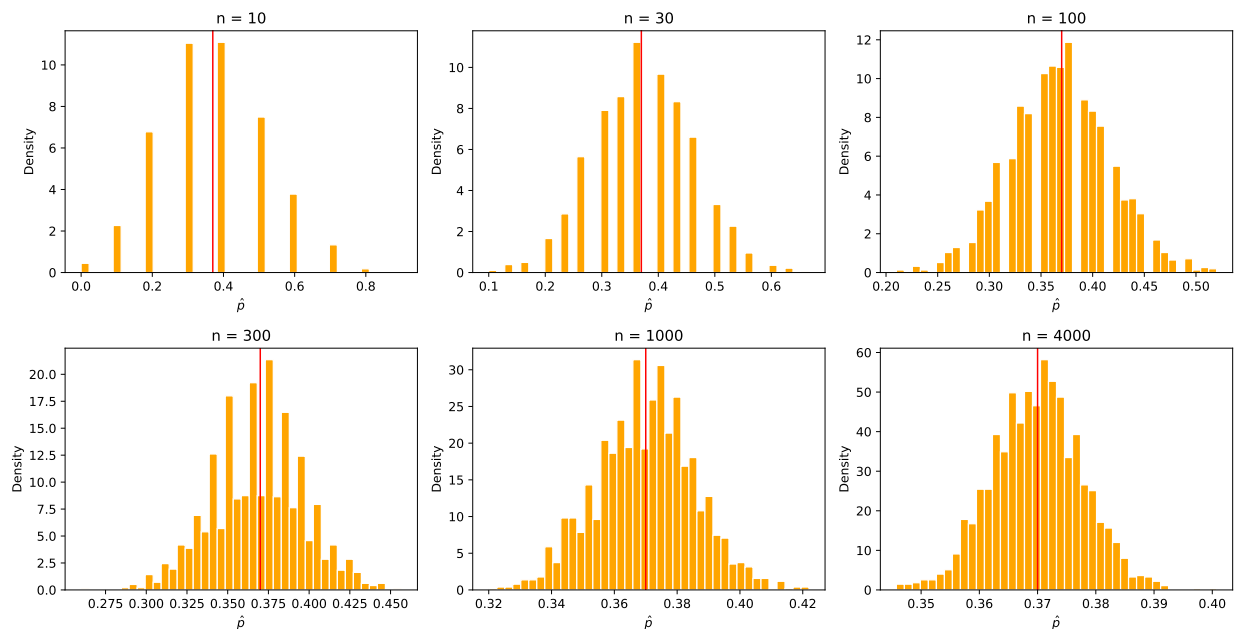
    ax.axvline(p_true, color="red", linewidth=1.2)
    ax.set_title(f"n = {n}")
    ax.set_xlabel(r"$\hat{p}$")
    ax.set_ylabel("Density")

fig.suptitle(
    "Consistency of the MLE for Bernoulli(p)\n"
    "As sample size increases, the sampling distribution of $\hat{p}$ concentrates near the true $p$",
    fontsize=16,
    y=1.02
)

fig.tight_layout()
plt.show()

```

As sample size increases, the sampling distribution of $\hat{p}$ concentrates near the true p



```

import numpy as np
import matplotlib.pyplot as plt

```

```

from scipy.stats import norm

np.random.seed(1)

p_true = 0.25
Nsim = 50000

# -----
# n = 50
# -----
n50 = 50
phat50 = np.array([np.mean(np.random.binomial(1, p_true, n50)) for _ in range(Nsim)])
sd50 = np.sqrt(p_true * (1 - p_true) / n50)

# -----
# n = 500
# -----
n500 = 500
phat500 = np.array([np.mean(np.random.binomial(1, p_true, n500)) for _ in range(Nsim)])
sd500 = np.sqrt(p_true * (1 - p_true) / n500)

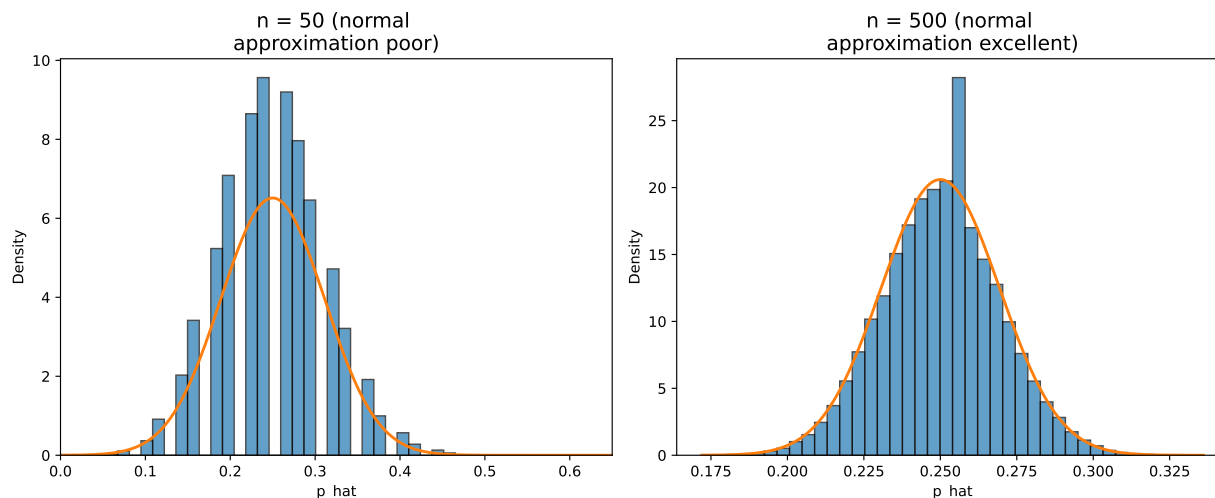
# -----
# Plot side-by-side
# -----
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Left: n = 50
axes[0].hist(phat50, bins=35, density=True, alpha=0.7, edgecolor="black")
x_vals50 = np.linspace(0, 0.65, 500)
axes[0].plot(x_vals50, norm.pdf(x_vals50, loc=p_true, scale=sd50), linewidth=2)
axes[0].set_xlim(0, 0.65)
axes[0].set_title("n = 50 (normal \napproximation poor)", fontsize=14)
axes[0].set_xlabel("p_hat")
axes[0].set_ylabel("Density")

# Right: n = 500
axes[1].hist(phat500, bins=40, density=True, alpha=0.7, edgecolor="black")
x_vals500 = np.linspace(phat500.min(), phat500.max(), 500)
axes[1].plot(x_vals500, norm.pdf(x_vals500, loc=p_true, scale=sd500), linewidth=2)
axes[1].set_title("n = 500 (normal \napproximation excellent)", fontsize=14)
axes[1].set_xlabel("p_hat")
axes[1].set_ylabel("Density")

plt.tight_layout()
plt.show()

```



```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

np.random.seed(2)

n = 50
p_true = 0.37
B = 2000

# Simulate MLEs
phats = np.array([np.mean(np.random.binomial(1, p_true, n)) for _ in range(B)])

# Plot histogram with density scaling
plt.figure(figsize=(7,5))
plt.hist(phats, bins=30, density=True, edgecolor='black', color='lightgray')

# Normal approximation overlay
x_vals = np.linspace(phats.min(), phats.max(), 500)
plt.plot(
    x_vals,
    norm.pdf(x_vals, loc=p_true, scale=np.sqrt(p_true * (1 - p_true) / n)),
    linestyle='dashed',
    linewidth=1
)

plt.title(f'Distribution of MLE (n={n}, p={p_true:.2f})')
plt.xlabel('phat')
plt.ylabel('density')
plt.tight_layout()
plt.show()
```

